Generating Chest Radiographs with a Convolutional Variational Autoencoder

ECE 285 Homework 1

Abstract

We train one convolutional variational autoencoder (VAE) on the Chest X-Ray Images (Pneumonia) training split after reserving a stratified validation subset, then draw unconditional samples from its Gaussian prior. We report training behavior, example radiographs, Inception Score (IS), and an explicitly identified 1000-feature Fréchet Inception Distance (FID-1000). The generated images capture coarse chest structure but blur fine anatomy; these natural-image metrics do not establish diagnostic quality.

1 Data and method

We use the Kaggle pediatric chest X-ray dataset [1]. The original training split is stratified into 4,694 training and 522 validation images (seed 285); the separate 624-image test split is reserved for evaluation. One VAE covers both NORMAL and PNEUMONIA images. Images are converted to grayscale, resized to 64×64, and scaled to $[0, 1]$.

Four stride-two convolutional encoder layers produce $4 \times 4 \times 128$ features. Linear heads parameterize a 32-dimensional diagonal Gaussian posterior; a reparameterized sample enters a linear layer and four transposed convolutions ending in a sigmoid. The model has 543,649 trainable parameters. Following Kingma and Welling [2], we minimize per-image pixelwise binary cross-entropy (summed over pixels) plus $\beta D_{\mathrm{KL}}[q_\phi(z \mid x) \| \mathcal{N}(0, I)]$, with $\beta = 1$. At generation time, $z \sim \mathcal{N}(0, I)$; no test image is supplied.

2 Training

We train with Adam (learning rate 10^{-3}), batch size 64, for 20 epochs on CPU. The checkpoint is selected by minimum validation negative ELBO. The test split does not enter training or checkpoint selection. Figure 1 shows convergence; Table 1 breaks down the selected epoch-20 losses.

3 Generative evaluation

We compare independent prior samples with randomly chosen official test images (seed 286). For Ignite's IS and FID implementation [3], each grayscale image is repeated to RGB, resized to 299×299, scaled to $[-1, 1]$, and evaluated by ImageNet-pretrained Inception-v3. Table 1 gives scores and sample counts. FID-1000 uses the model's 1000-class probabilities as features, as in the cited Ignite tutorial; it is not comparable to the usual pool3/2048-feature FID [4].

Preprint.

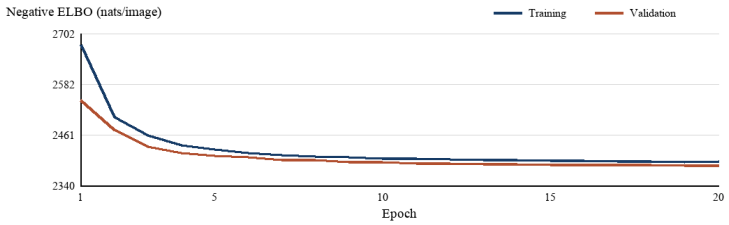
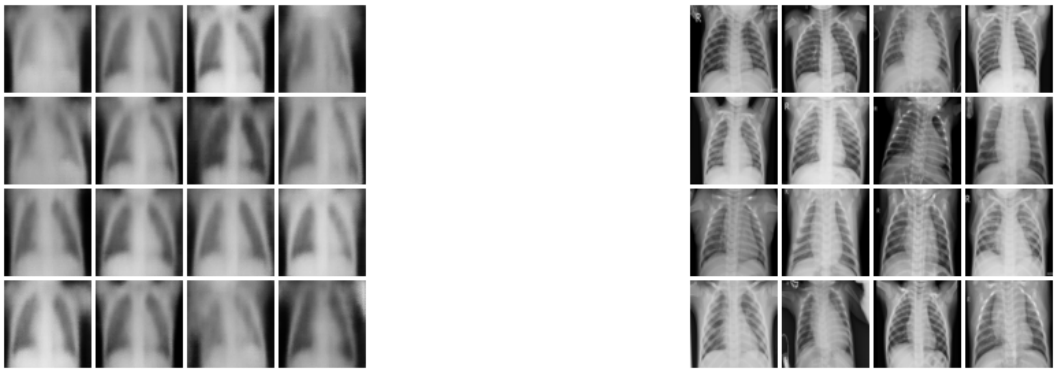


Figure 1: Mean negative ELBO in nats/image on training and validation splits; lower is better. The selected checkpoint is epoch 20.



(a) Prior samples (b) Held-out test images

Figure 2: Generated and real 64×64 grayscale radiographs displayed at the same scale. The model recovers coarse chest outlines but loses rib and lung texture.

Table 1: Measured results at the selected epoch 20. Losses are per image; "gen/test" denotes generated and held-out images.

Loss (nats/image)	Training	Validation
Reconstruction BCE	2,375.02	2,365.96
KL divergence	22.50	22.14
Negative ELBO	2,397.53	2,388.10
Generative metric	Score	Images (gen/test)
IS	1.779	256 / –
FID-1000	0.177	256 / 256

4 Discussion

Figure 2 shows visibly smoother generated images than real test images. Downsampling removes fine lung detail, and pixelwise binary cross-entropy with a compact Gaussian latent model can favor smooth outputs. The training set is class-imbalanced; this unconditional model cannot select pathology. The small-sample IS and FID-1000 do not establish clinical realism or diagnostic performance. Higher resolution and radiology-specific evaluation would be useful next steps.

Code and reproducibility: https://github.com/Tobudase/ece285-hw1-chest-xray-vae.

References

[1] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle, 2018. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.

[2] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.

[3] PyTorch-Ignite. GAN evaluation using FID and IS. https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/.

[4] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS, 2017.